Total No. of Questions : 8] [Total No. of Printed Pages : 3

Paper Code : 21327

F-427

B.C.A. (Sixth Semester) Examination, 2023

(New Course)

Paper No. BCA-N-602

ARTIFICIAL INTELLIGENCE

Time : Three Hours] *[Maximum Marks : 70*

Note : Attempt any **five** questions. **All** questions carry equal marks. Symbols used have their usual meanings.

1. (a) What is 'AI'? Explain its importance in today's scenario with proper examples.

   (b) Differentiate heuristic algorithm and solution guaranteed algorithm.

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2. (a) Describe the parsing techniques.

   (b) Write in brief about :

   (i) Fillmore's Grammers

   (ii) Context free grammer

3. (a) What do you mean by "First order logic"? Write one example.

   (b) What is the difference between forward deduction and backward deduction?

4. (a) Define "PROLOG" with its application. What are miniskey frames?

   (b) How "MYCIN" expert system is different from "DENDRAL" expert system?

5. (a) Compare "NLP' with 'expert' system giving their applications.

   (b) Write down syntax of both 'LISP' and 'PROLOG'. Explain the applications of LISP in programming world.

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6. (a) How pattern-recognition is implemented using the concept of artificial intelligence?

(b) Compare various techniques used for knowledge representation.

7. (a) Explain transition nots and augmented transitions nots with suitable examples.

(b) Illustrate the "Horn-Clause" with two examples.

8. Write short notes on : (any **four**)

(i) Meta knowledge

(ii) Applications of AI in education

(iii) Speech recognition

(iv) Shanks conceptual dependency

(v) CFG

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